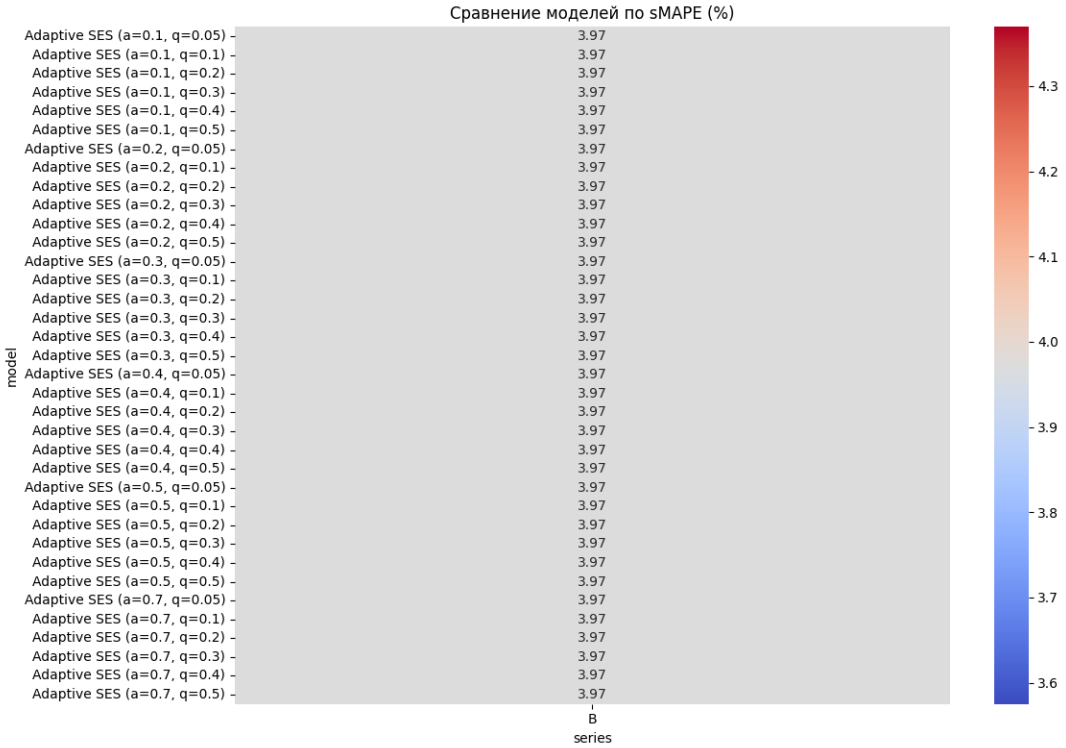
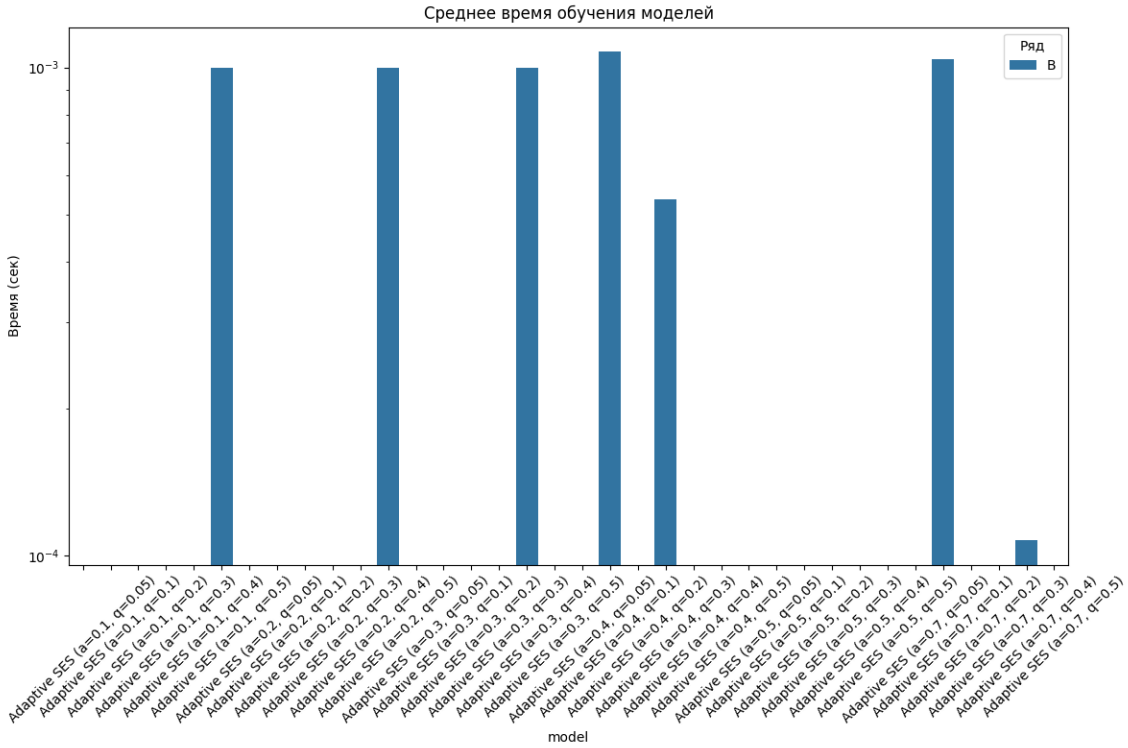
Полный отчет анализа моделей

Сводные результаты

Серия	Модель	Средняя MAE	Средняя RMSE	Средняя sMAPE	Среднее время обучения	Лучшая модель
B	Adaptive SES (a=0.1, q=0.05)	3.9970	5.1590	3.9722%	0.00 сек	+
B	Adaptive SES (a=0.1, q=0.1)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.1, q=0.2)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.1, q=0.3)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.1, q=0.4)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.1, q=0.5)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.2, q=0.05)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.2, q=0.1)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.2, q=0.2)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.2, q=0.3)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.2, q=0.4)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.2, q=0.5)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.3, q=0.05)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.3, q=0.1)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.3, q=0.2)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.3, q=0.3)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.3, q=0.4)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.3, q=0.5)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.4, q=0.05)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.4, q=0.1)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.4, q=0.2)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.4, q=0.3)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.4, q=0.4)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.4, q=0.5)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.5, q=0.05)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.5, q=0.1)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.5, q=0.2)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.5, q=0.3)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.5, q=0.4)	3.9970	5.1590	3.9722%	0.00 сек	

B	Adaptive SES (a=0.5, q=0.5)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.7, q=0.05)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.7, q=0.1)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.7, q=0.2)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.7, q=0.3)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.7, q=0.4)	3.9970	5.1590	3.9722%	0.00 сек	
B	Adaptive SES (a=0.7, q=0.5)	3.9970	5.1590	3.9722%	0.00 сек	




Среднее время обучения моделей
Ряд
B
10^{-3}
10^{-4}
Время (сек)
Adaptive SES (a=0.1, q=0.05)
Adaptive SES (a=0.1, q=0.1)
Adaptive SES (a=0.1, q=0.2)
Adaptive SES (a=0.1, q=0.3)
Adaptive SES (a=0.1, q=0.4)
Adaptive SES (a=0.1, q=0.5)
Adaptive SES (a=0.2, q=0.05)
Adaptive SES (a=0.2, q=0.1)
Adaptive SES (a=0.2, q=0.2)
Adaptive SES (a=0.2, q=0.3)
Adaptive SES (a=0.2, q=0.4)
Adaptive SES (a=0.2, q=0.5)
Adaptive SES (a=0.3, q=0.05)
Adaptive SES (a=0.3, q=0.1)
Adaptive SES (a=0.3, q=0.2)
Adaptive SES (a=0.3, q=0.3)
Adaptive SES (a=0.3, q=0.4)
Adaptive SES (a=0.3, q=0.5)
Adaptive SES (a=0.4, q=0.05)
Adaptive SES (a=0.4, q=0.1)
Adaptive SES (a=0.4, q=0.2)
Adaptive SES (a=0.4, q=0.3)
Adaptive SES (a=0.4, q=0.4)
Adaptive SES (a=0.4, q=0.5)
Adaptive SES (a=0.5, q=0.05)
Adaptive SES (a=0.5, q=0.1)
Adaptive SES (a=0.5, q=0.2)
Adaptive SES (a=0.5, q=0.3)
Adaptive SES (a=0.5, q=0.4)
Adaptive SES (a=0.5, q=0.5)
Adaptive SES (a=0.7, q=0.05)
Adaptive SES (a=0.7, q=0.1)
Adaptive SES (a=0.7, q=0.2)
Adaptive SES (a=0.7, q=0.3)
Adaptive SES (a=0.7, q=0.4)
Adaptive SES (a=0.7, q=0.5)
model

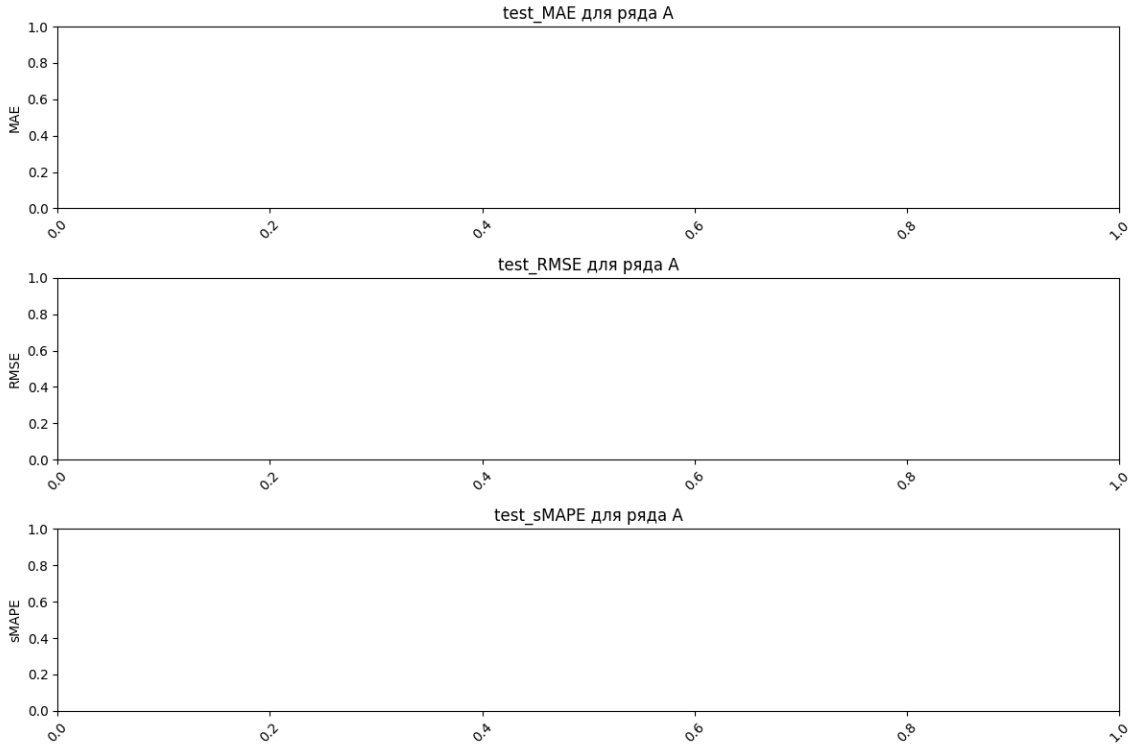
Анализ ряда A

Лучшая модель:

0
A
GRU (16u, 24w, 150e), 16b)
11.704305171966553
0.3123993601078594
0.33180706786782527
12.217951856981255
0.5437099530568105
0.5643125003292365
20.320353423394945
False

Детали модели:

Лучшая модель для ряда A: GRU (16u, 24w, 150e), 16b)
Тестовые метрики:
MAE: 0.5437
RMSE: 0.5643
sMAPE: 20.3204%
Время обучения: 11.70 сек


test_MAE для ряда A
MAE
test_RMSE для ряда A
RMSE
test_sMAPE для ряда A
sMAPE
0.0
0.2
0.4
0.6
0.8
1.0

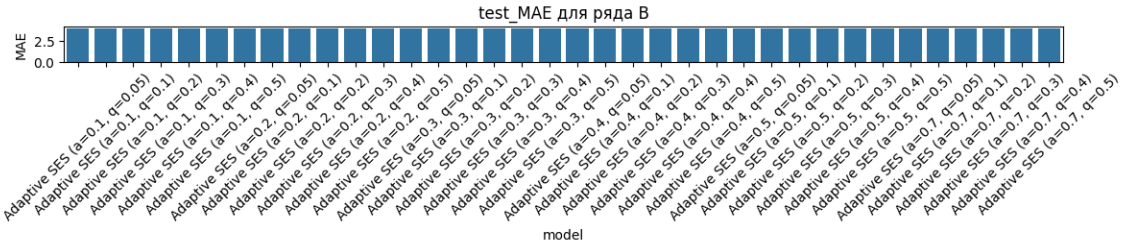
Анализ ряда B

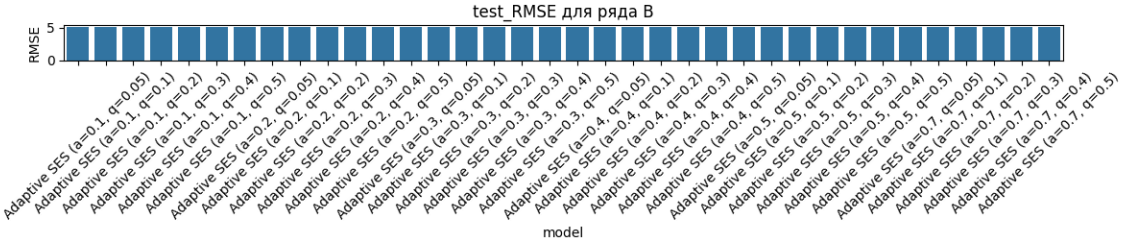
Лучшая модель:

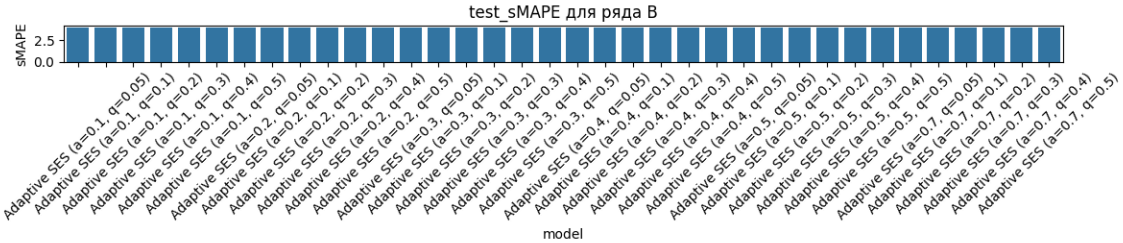
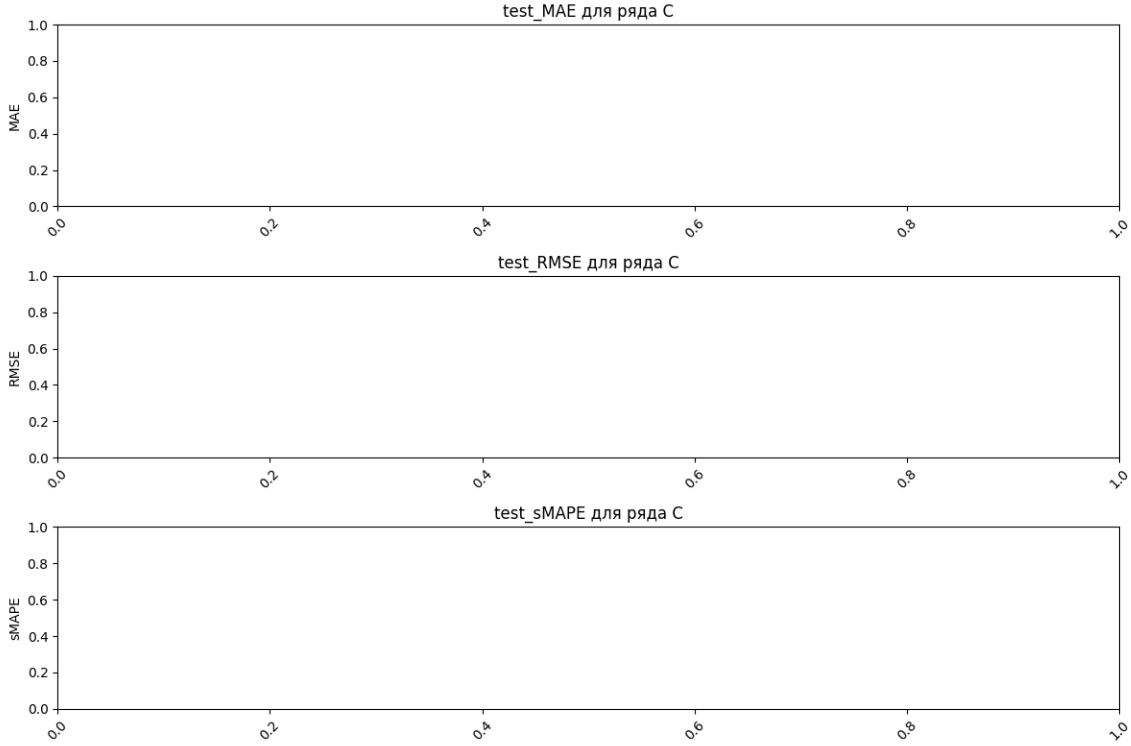
0
B
Adaptive SES (a=0.1, q=0.05)
0.0
4.453125
5.5049693459637
4.424284344684243
3.9969696969696953
5.158958081323056
3.972161277592238
False

Детали модели:

Лучшая модель для ряда B: Adaptive SES (a=0.1, q=0.05)
Тестовые метрики:
MAE: 3.9970
RMSE: 5.1590
sMAPE: 3.9722%
Время обучения: 0.00 сек


test_MAE для ряда B
MAE
2.5
0.0
Adaptive SES (a=0.1, q=0.05)
Adaptive SES (a=0.1, q=0.1)
Adaptive SES (a=0.1, q=0.2)
Adaptive SES (a=0.1, q=0.3)
Adaptive SES (a=0.1, q=0.4)
Adaptive SES (a=0.1, q=0.5)
Adaptive SES (a=0.2, q=0.05)
Adaptive SES (a=0.2, q=0.1)
Adaptive SES (a=0.2, q=0.2)
Adaptive SES (a=0.2, q=0.3)
Adaptive SES (a=0.2, q=0.4)
Adaptive SES (a=0.2, q=0.5)
Adaptive SES (a=0.3, q=0.05)
Adaptive SES (a=0.3, q=0.1)
Adaptive SES (a=0.3, q=0.2)
Adaptive SES (a=0.3, q=0.3)
Adaptive SES (a=0.3, q=0.4)
Adaptive SES (a=0.3, q=0.5)
Adaptive SES (a=0.4, q=0.05)
Adaptive SES (a=0.4, q=0.1)
Adaptive SES (a=0.4, q=0.2)
Adaptive SES (a=0.4, q=0.3)
Adaptive SES (a=0.4, q=0.4)
Adaptive SES (a=0.4, q=0.5)
Adaptive SES (a=0.5, q=0.05)
Adaptive SES (a=0.5, q=0.1)
Adaptive SES (a=0.5, q=0.2)
Adaptive SES (a=0.5, q=0.3)
Adaptive SES (a=0.5, q=0.4)
Adaptive SES (a=0.5, q=0.5)
Adaptive SES (a=0.7, q=0.05)
Adaptive SES (a=0.7, q=0.1)
Adaptive SES (a=0.7, q=0.2)
Adaptive SES (a=0.7, q=0.3)
Adaptive SES (a=0.7, q=0.4)
Adaptive SES (a=0.7, q=0.5)
model


test_RMSE для ряда B
RMSE
5
0
Adaptive SES (a=0.1, q=0.05)
Adaptive SES (a=0.1, q=0.1)
Adaptive SES (a=0.1, q=0.2)
Adaptive SES (a=0.1, q=0.3)
Adaptive SES (a=0.1, q=0.4)
Adaptive SES (a=0.1, q=0.5)
Adaptive SES (a=0.2, q=0.05)
Adaptive SES (a=0.2, q=0.1)
Adaptive SES (a=0.2, q=0.2)
Adaptive SES (a=0.2, q=0.3)
Adaptive SES (a=0.2, q=0.4)
Adaptive SES (a=0.2, q=0.5)
Adaptive SES (a=0.3, q=0.05)
Adaptive SES (a=0.3, q=0.1)
Adaptive SES (a=0.3, q=0.2)
Adaptive SES (a=0.3, q=0.3)
Adaptive SES (a=0.3, q=0.4)
Adaptive SES (a=0.3, q=0.5)
Adaptive SES (a=0.4, q=0.05)
Adaptive SES (a=0.4, q=0.1)
Adaptive SES (a=0.4, q=0.2)
Adaptive SES (a=0.4, q=0.3)
Adaptive SES (a=0.4, q=0.4)
Adaptive SES (a=0.4, q=0.5)
Adaptive SES (a=0.5, q=0.05)
Adaptive SES (a=0.5, q=0.1)
Adaptive SES (a=0.5, q=0.2)
Adaptive SES (a=0.5, q=0.3)
Adaptive SES (a=0.5, q=0.4)
Adaptive SES (a=0.5, q=0.5)
Adaptive SES (a=0.7, q=0.05)
Adaptive SES (a=0.7, q=0.1)
Adaptive SES (a=0.7, q=0.2)
Adaptive SES (a=0.7, q=0.3)
Adaptive SES (a=0.7, q=0.4)
Adaptive SES (a=0.7, q=0.5)
model


test_sMAPE для ряда B
sMAPE
2.5
0.0
Adaptive SES (a=0.1, q=0.05)
Adaptive SES (a=0.1, q=0.1)
Adaptive SES (a=0.1, q=0.2)
Adaptive SES (a=0.1, q=0.3)
Adaptive SES (a=0.1, q=0.4)
Adaptive SES (a=0.1, q=0.5)
Adaptive SES (a=0.2, q=0.05)
Adaptive SES (a=0.2, q=0.1)
Adaptive SES (a=0.2, q=0.2)
Adaptive SES (a=0.2, q=0.3)
Adaptive SES (a=0.2, q=0.4)
Adaptive SES (a=0.2, q=0.5)
Adaptive SES (a=0.3, q=0.05)
Adaptive SES (a=0.3, q=0.1)
Adaptive SES (a=0.3, q=0.2)
Adaptive SES (a=0.3, q=0.3)
Adaptive SES (a=0.3, q=0.4)
Adaptive SES (a=0.3, q=0.5)
Adaptive SES (a=0.4, q=0.05)
Adaptive SES (a=0.4, q=0.1)
Adaptive SES (a=0.4, q=0.2)
Adaptive SES (a=0.4, q=0.3)
Adaptive SES (a=0.4, q=0.4)
Adaptive SES (a=0.4, q=0.5)
Adaptive SES (a=0.5, q=0.05)
Adaptive SES (a=0.5, q=0.1)
Adaptive SES (a=0.5, q=0.2)
Adaptive SES (a=0.5, q=0.3)
Adaptive SES (a=0.5, q=0.4)
Adaptive SES (a=0.5, q=0.5)
Adaptive SES (a=0.7, q=0.05)
Adaptive SES (a=0.7, q=0.1)
Adaptive SES (a=0.7, q=0.2)
Adaptive SES (a=0.7, q=0.3)
Adaptive SES (a=0.7, q=0.4)
Adaptive SES (a=0.7, q=0.5)
model

Анализ ряда C